

Impact of Model Complexity on Transfer Learning-Based Pneumonia Detection Using Public Chest X-Ray Images

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Abstract—This paper examines how architectural complexity impacts the effectiveness of the transfer learning models in automated pneumonia detection by using the chest X-ray images. The experiments were conducted using the publicly available Chest X-Ray Images (Pneumonia) dataset from Kaggle [8]. Three convolutional neural network models, which are MobileNetV2, ResNet50 and EfficientnetB0, were tested on a standardised training pipeline of transfer learning. Each of the models was pre-trained on ImageNet and converted to binary classification. According to the experimental findings, the lightweight MobileNetV2 model performed better in the generalisation of the results, with the test accuracy of 89.9% and the receiver operating characteristic area under the curve (ROC-AUC) of 0.972. Conversely, more architectural layers revealed more sensitivity to the imbalance of the classes and lower stability in training. These results indicate that the complexity of the building does not always correlate with the enhancement of the diagnostic performance, especially with small and unbalanced medical imaging datasets.

Index Terms—Pneumonia Detection, Transfer Learning, Deep Learning, Medical Imaging, Model Complexity, Chest X-Ray

I. INTRODUCTION

Pneumonia has been a major world health burden in that it has been a considerable cause of morbidity and mortality in both pediatric and adult population groups. The timely and precise diagnosis is one of the key factors to guarantee timely clinical intervention and limit negative outcomes. The most common diagnostic methods of pneumonia detection include chest radiography; the former is time-consuming, and it is prone to inter-observer differences when the chest X-ray images are manually interpreted.

The recent progress in the field of deep learning has shown promising results in automated medical image analysis tasks. In particular, convolutional neural networks (CNNs) have shown strong capability in extracting hierarchical image features for disease classification [1], [5]. Applications of transfer learning, in which models trained on large-scale datasets (ImageNet) are transferred to the medical imaging task, have become the most popular method of pneumonia detection because annotated medical datasets are often not available in large quantities [7], [8].

Although numerous studies report high performance using deep CNN architectures [3], [10], the relationship between model complexity and generalisation performance in pneumonia detection remains insufficiently examined. Surpassing the performance of larger architectures with more parameters might not always be achieved, especially with imbalanced or small datasets and certainly not only through large datasets and simulations [2], [15].

This paper examines how the complexity of architecture affects transfer learning based pneumonia detection utilising publicly available chest X-ray data as a condition. Three pre-trained models, MobileNetV2, ResNet50 and EfficientNetB0 are tested in a standardised training pipeline to determine variations in performance, computational cost, and imbalance resistance to classes.

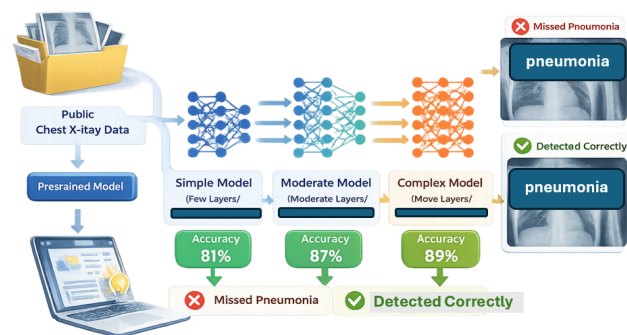


Fig. 1. Proof of Concept

II. RELATED WORK

The recent years research on deep learning methods of detecting pneumonia has been well researched. Comprehensive surveys highlight the rapid evolution of CNN-based architectures and transfer learning strategies for chest X-ray analysis [1], [9]. There are several studies that report good performance of the pretrained models fine-tuned on the pneumonia datasets.

Transfer learning has been widely adopted due to limited dataset sizes in medical imaging [7], [8]. Salehi et al. [3] demonstrated effective pneumonia classification using deep transfer learning on pediatric chest X-rays, while Manickam et al. [10] compared multiple pretrained architectures for automated detection. Ensemble approaches have also been proposed to improve robustness [6].

Recent work has also explored architectural variations, attention mechanisms, and hyperparameter customisation to enhance performance [12], [16]. However, several studies note that model performance can degrade under class imbalance conditions, and imbalance mitigation strategies are often necessary [2], [15].

In spite of these developments, there has been little comparative experimentation in systematically assessing the effect that architectural complexity has on generalisation performance in a standardised experimental setting. This paper seeks to give a controlled comparison between lightweight and deeper transfer learning models with a view to understand the behaviour of these models better when it comes to pneumonia detection tasks.

In recent years, the analysis of medical images by deep learning approaches has been highly developed to analyse automated medical images. Convolutional neural networks (CNNs) have proven highly effective in detection of pathological images in radiological images such as chest X-rays, computed tomography as well as magnetic resonance imaging data. Several studies have reported that deep CNN architectures can achieve performance comparable to that of experienced radiologists in certain diagnostic tasks [5]. The models are capable of automatically learning hierarchical feature representations on raw image data and eliminates the requirement of human engineered feature engineering.

With small quantities of annotated medical images, transfer learning has emerged as a popular approach to medical image classification. Pretrained models (VGGNet, ResNet, MobileNet, and EfficientNet) are trained on domain-specific data instead of training networks. This approach allows models to leverage previously learned visual representations from large-scale datasets such as ImageNet, improving training efficiency and generalisation performance [7]. Several studies have successfully applied transfer learning for pneumonia detection using chest X-ray datasets, reporting high classification accuracy and improved diagnostic support for clinical environments [3], [10].

Regardless of these developments, the connexion between the complexity of a model and the classification performance of medical imaging tasks are still a subject of active research. The bigger architectures tend to have much more parameters, which leads to higher computational costs and hence overfitting on fairly small datasets. Some studies have suggested that lightweight architectures such as MobileNet may offer competitive performance while maintaining lower computational cost [14]. Other works have highlighted the importance of addressing class imbalance in medical datasets, as imbalanced training distributions can significantly affect

model reliability and diagnostic sensitivity [2], [15].

Hence, a comparative study in terms of lightweight and deeper architectures operating in the same experimental conditions should be conducted in a systematic manner in order to gain a deeper insight into the effects of architectural complexity on generalisation of models in pneumonia detection tasks.

III. METHODOLOGY

A. Conceptual Framework

The theoretical design of the suggested pneumonia detecting system shows the connection between data set preparation, transfer learning models, training process, and performance assessment. The framework starts with the distribution of the chest X-ray images by open-source data. The images are pre-processed and augmented in order to enhance generalisation in the course of training.

After preprocessing, pretrained convolutional neural network architectures are modified by using transfer learning. Standardised training configuration is used to train the models and classification metrics such as accuracy, sensitivity, specificity, and ROC-AUC, are used to evaluate the models. The overall workflow of the proposed system is illustrated in Fig. 2.

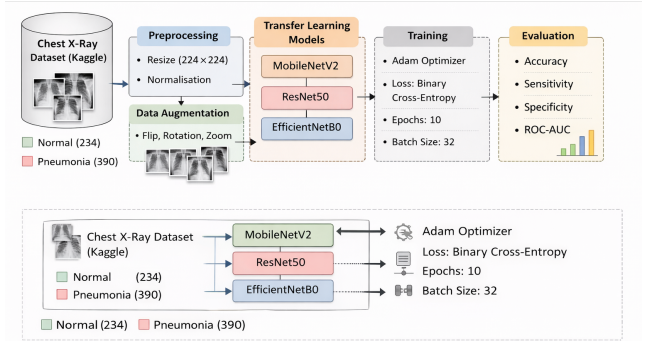


Fig. 2. Conceptual framework of the proposed transfer learning-based pneumonia detection pipeline.

B. Dataset Description

In this work, the publicly available Chest X-Ray Images (Pneumonia) dataset was used, which was acquired through Kaggle. The data set contains anterior-posterior chest radiographs categorised into two classes: *Normal* and *Pneumonia*. Public chest X-ray datasets are widely used in pneumonia detection research due to their accessibility and reproducibility [1], [8].

The established training and testing divisions were utilised. The original training set was further divided into training (85%) and validation (15%) subsets using an internal validation split to ensure consistent comparison across models. The test set was 624 images comprising of 234 normal and 390 cases of pneumonia. The overall experimental framework is illustrated in Fig. 3.

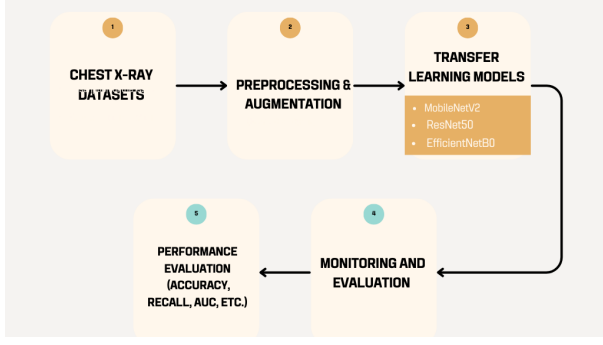


Fig. 3. Proposed transfer learning-based pneumonia detection framework.

C. Preprocessing and Data Augmentation

All images were resized to 224×224 pixels to match the input requirements of the pretrained architectures. For MobileNetV2 and ResNet50 experiments, pixel values were normalized to the range $[0,1]$. In the case of EfficientNetB0, the architecture-specific preprocessing task was used to scale the input to the pretraining criteria of ImageNet.

To minimize overfitting and enhance the generalization, the training data has been data augmented using horizontal flipping, small rotations (possibly up to 10 degrees), and zoom transformations. Validation and test sets were not augmented.

D. Transfer Learning Architectures

Transfer learning has shown strong performance in medical image classification, particularly when annotated datasets are limited [7], [10]. The three pretrained convolutional neural network models were tested:

- MobileNetV2
- ResNet50
- EfficientNetB0

ImageNet pretrained weights were used to initialize all the models. The original classification head was eliminated and it was replaced with a custom classification head that comprised of:

- Global Average Pooling layer
- Fully connected dense layer (128 units, ReLU activation)
- Dropout layer (0.3)
- Sigmoid output layer

Each network was trained to freeze the convolutional base so that it is possible to compare fair and maintain the pre-trained feature representations. The generic model design used for all architectures is illustrated in Fig. 4.

E. Training Configuration

All models were trained using the Adam optimizer with a learning rate of 1×10^{-4} and binary cross-entropy loss. The training was done in 10 epochs and batch size was 32.

Given the class imbalance in the dataset, additional experiments were conducted for EfficientNetB0 using class weighting strategies to investigate imbalance mitigation effects [2], [15]. The process of the training was conducted in Google

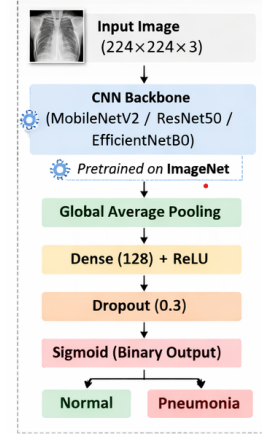


Fig. 4. Transfer learning model architecture used for pneumonia detection.

Colab using a GPU acceleration. Complexity Analysis parameters and training time were recorded.

F. Proposed Algorithm

Algorithm 1 Transfer Learning-Based Pneumonia Detection Framework

- 1: Load chest X-ray dataset
- 2: Split dataset into training (85%), validation (15%), and test sets
- 3: Resize images to 224×224 pixels
- 4: Apply data augmentation to training data
- 5: **for** each pretrained model in {MobileNetV2, ResNet50, EfficientNetB0} **do**
- 6: Initialize model with ImageNet weights
- 7: Replace classification head with custom binary classifier
- 8: Freeze convolutional base layers
- 9: Train model using Adam optimizer and binary cross-entropy loss
- 10: Evaluate model on test set
- 11: **end for**
- 12: Compute Accuracy, Sensitivity, Specificity, and ROC-AUC

G. Evaluation Metrics

Independent test set was assessed using the following metrics on performance:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic – Area Under Curve (ROC-AUC)
- Confusion Matrix

Pneumonia detection recall was especially important because of the clinical implications of the research process [4], [5].

IV. RESULTS AND DISCUSSION

The comparative performance of the evaluated architectures is summarised in Table I. A detailed classification report of the evaluated models is presented in Table II, summarising precision, recall, F1-score, and accuracy metrics.

		MobileNetV2 (Acc = 89.9%)		ResNet50 (Acc = 64.3%)		EfficientNetB0 (Acc = 62.3%)	
		Normal	Pneumonia	Normal	Pneumonia	Normal	Pneumonia
Actual	Normal	188 (TN)	46 (FP)	19 (TN)	215 (FP)	59 (TN)	175 (FP)
	Pneumonia	16 (FN)	374 (TP)	8 (FN)	382 (TP)	59 (FN)	331 (TP)
		Normal	Pneumonia	Normal	Pneumonia	Normal	Pneumonia
		Predicted		Predicted		Predicted	

Fig. 5. Confusion matrices of the evaluated models on the test dataset. The matrices illustrate the distribution of true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP) for MobileNetV2, ResNet50, and EfficientNetB0.

TABLE I
COMPARATIVE PERFORMANCE OF TRANSFER LEARNING MODELS

Model	Params	Test Acc	ROC-AUC	Sensitivity	Specificity
MobileNetV2	2.42M	89.9%	0.972	0.96	0.80
ResNet50	23.85M	64.3%	0.838	0.98	0.08
EfficientNetB0	4.21M	62.3%	0.665	0.85	0.25

TABLE II
CLASSIFICATION PERFORMANCE OF EVALUATED TRANSFER LEARNING MODELS

Model	Precision	Recall	F1-score	Accuracy
MobileNetV2	0.89	0.96	0.92	0.899
ResNet50	0.64	0.98	0.77	0.643
EfficientNetB0	0.65	0.85	0.74	0.623

Specificity was computed as $TN/(TN + FP)$ with the Normal class treated as the negative class, while sensitivity corresponds to pneumonia recall.

Clinically, high recall in pneumonia detection is most important to reduce occurrence of false negative but excessive reduction in specificity, as observed in the ResNet50 model, could hasten unnecessary clinical follow-up intervention. Therefore, it is still essential to achieve a reasonable balance between sensitivity and specificity to make a successful real-world implementation.

In addition to predictive performance, computational efficiency was also taken into account on the basis of the number of parameters and time used to train architectures.

MobileNetV2 had the highest test accuracy (89.9% and ROC-AUC of 0.972), even though the number of parameters is the lowest (2.42 million). Pneumonia recall (0.96) was high and normal cases were balanced with the model. Such results mean that there is effective generalisation and computational efficiency.

The ResNet50, which has around 23.85 million parameters, has a test accuracy of 64.3% and ROC-AUC value of 0.838.

Whereas, pneumonia recall was high (0.98), the model incorrectly classified a very large proportion of normal cases as pneumonia, which is poor specificity. This indicates that the more architectural depth was added, the less generalisation performance was achieved in this dataset.

EfficientNetB0 was originally found to be unstable in the situation of class imbalance, falling to single-class predictors with no imbalance alleviation. Once architecture-specific pre-processing and class weighting were implemented, the performance stabilised and the test accuracy was 62.3% and ROC-AUC of 0.665. Nevertheless, a high generalisation difference was found between validation results and test results which suggests sensitivity to the imbalance in databases and possible distribution changes.

All in all, the findings indicate that the enhanced complexity of the architecture does not always provide better performance to pneumonia detection with limited and unbalanced publicly available datasets. In more real-world applications, lightweight neural networks like MobileNetV2 can be a better trade-off between computational and predictive efficiency.

A. Computational Complexity Analysis

Besides the predictive performance, computational efficiency was also investigated through the comparison of the parameter counts and training times. The number of parameters in MobileNetV2 was about 2.42 million, much lower than the number of parameters in the ResNet50 (23.85 million) and EfficientNetB0 (4.21 million). Even though it is less complex, MobileNetV2 had better generalisation.

This suggests that the performance of limited-data medical imaging situations does not necessarily get better with increased architectural depth and use of more parameters. Lightweight architectures can offer a more realistic tradeoff between the cost and predictive performance of the computation.

V. CONCLUSION

This paper has investigated the effects of the model complexity when detecting pneumonia through transfer learning with a publicly available chest X-ray dataset. Three ready-made CNN models, namely MobileNetV2, ResNet50 and EfficientNetB0, were tested in a standardised training system.

It is experimentally confirmed that the lightweight MobileNetV2 architecture was also better in its ability to generalise than more complex and deep networks. In larger architectures, there was a decrease in stability and an increase in sensitivity to class imbalance, and it is important to note that model selection is important when dealing with small medical imaging datasets.

Limitations

A single publicly available dataset was used in this study, possibly not a representative of clinical variability in real life. Also the backbone layers were frozen so that they can be compared; more performance can be obtained by performing selective fine-tuning. Further research must take into account

the cross-dataset validation and multi-institutional testing in order to test a bigger generalisability.

Future research can be done on more imbalance mitigating methods, selective fine-tuning of backbone layers, and testing on larger multi-institutional datasets to further confirm these findings.

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